Questions of the day

- What's the best way to store a graph in memory?
- How do we analyze algorithms using graphs?
- How should we iterate through the vertices and edges of a graph to solve graph problems?

Graph algorithms

William Hendrix

Lecture 9

Outline

- Review
  - Dynamic programming
  - Greedy algorithms
- Graph theory
- Graph representations
- Graph traversals

Dynamic programming review

- Design strategy for recursive problems with repeated subproblems

1. Solve problem recursively
   - Recurrence and base cases
2. Determine data structure
   - Usually based on number of changing parameters
   - E.g., LCS: start index of each string => 2D array

<u>Memoization</u>

3. Determine sentinel value
   - Not a valid solution
4. Wrapper function, memo check, store before returning

<u>Iterative dynamic programming</u>

3. Determine iteration order
   - Start at base case, move in opposition to recursion
4. Decide if space complexity can be reduced
5. Allocate data structure, write loops, recursion and return => reads and writes, problem variables => loop variables, return answer

Greedy algorithms review

- **Strategy:**
  - Break down problem into sequence of decisions
  - Make "best" choice for each decision
- **Example:** find largest subset with sum below a threshold
    - Choose min element
    - *Greedy idea:* min element gives more "room" to find other elements
    - Repeat until next element would exceed threshold
- Proof of correctness is tricky
  - *Intuition:* greedy is only correct when greedy choice always "as least as good" as any alternative
  - Many greedy algorithms are incorrect

Greedy algorithm exercise

- Prove that the greedy algorithms below do not find the largest **sum** $\leq t$ in a given array

```
Input: data: array of positive integers
Input: n: size of data
Input: t: target value
Output: subset of data with largest sum ≤ t
Algorithm: GreedyKnapsack
soln = {}
repeat
    Let m be the smallest element we haven't added yet
    Add m to soln if it doesn't push the sum over t
until the next smallest element is too big
return soln
```

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until the next smallest element is too big
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Counterexample:
data = [1, 10], t = 10

Optimal sol'n: [10]
Greedy answer: [1]

```
Input: data: array of positive integers
Input: n: size of data
Input: t: target value
Output: subset of data with largest sum ≤ t
Algorithm: GreedyKnapsack
soln = {}
repeat
    Let m be the largest element we haven't tried adding yet
    Add m to soln if it doesn't push the sum over t
until we can't add anything else
return soln
```

Counterexample:
data = [4, 3, 2], t = 5

Optimal sol'n: [3, 2]
Greedy answer: [4]

Greedy analysis

Pros

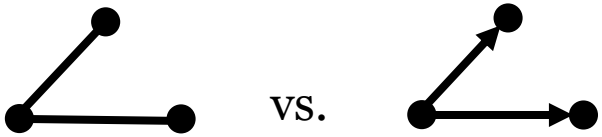
- Natural choice for *optimization problems*
  - Find the best/worst/biggest/smallest/etc.
- Relatively easy to design
- Usually very efficient
  - Heaps!
- Can often find "good enough" solution efficiently even if not correct
  - *Heuristic* technique

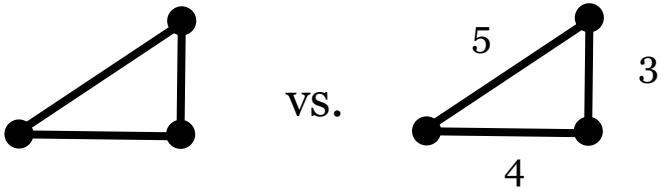
Cons

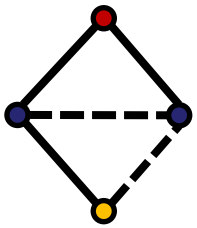
- Hard to prove correctness
- Often not correct

Graph theory review

- Mathematical abstraction for network of relationships
- **Vertices (nodes):** set of objects
  - Denoted as *V*
- **Edges:** set of relationships between vertices
  - Connect *two* vertices together
  - Denoted as *E*
- Graph variants
  - **Simple** graphs vs. multigraphs
    - No self loops
    - No edges between same pair of vertices
  - Directed vs. **undirected**
    - Symmetric or asymmetric relationships
  - Weighted vs. **unweighted**
    - Edges have "length" or "strength"
  - Labelled vs. **unlabelled**
    - Vertices and/or edges may have categories







Graph theory review

- **Order:** number of vertices in the graph
  - Usually denoted as n (or $|V|$)
- **Size:** number of edges in the graph
  - Usually represented as m (or $|E|$)
  - Graph complexity can depend on both n and m
    - $m = O(n^2)$
- **Adjacent:** two vertices with an edge between them
- **Incident:** vertex and an edge where edge connects to vertex
- **Degree:** number of vertices adjacent to a given vertex
  - Denoted $\deg(v)$ or $\deg_G(v)$
  - Max degree: denoted as Δ or $\Delta(G)$
  - Min degree: denoted as δ or $\delta(G)$
- **Neighborhood:** set of vertices adjacent to a given vertex
  - Denoted $\mathrm{N}(v)$ or $\mathrm{N}_G(v)$

Handshaking Lemma

Theorem 1. *For any simple graph $G = (V, E)$,*

$$\sum_{v \in V} \deg(v) = 2|E|,$$

where $\deg(v)$ *is the degree of vertex* v *in graph* G.

Proof (informal). When $m = 0$, every vertex has degree 0, so
degree sum $= 0 = 2m$.

Suppose true for graphs with k edges, and
let G have $k + 1$ edges.

Remove an edge (u, v)
$\Rightarrow$ new graph has k edges
$\Rightarrow$ new graph has degree sum $= 2k$.

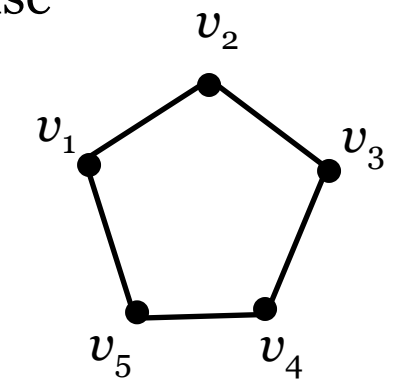
Degree of u and v one more in G, otherwise the same

G has degree sum $2k + 2 = 2m$.

Graph representation

- Three main representations of a graph in memory
  - C++: look up Boost Graph Library (www.boost.org)
- **Adjacency matrix**
  - n by n matrix
  - a_{ij} is 1 if v_i and v_j are adjacent, 0 otherwise

v_1	0	1	0	0	1
v_2	1	0	1	0	0
v_3	0	1	0	1	0
v_4	0	0	1	0	1
v_5	1	0	0	1	0



    - Symmetric for undirected graphs
      - *Optimization:* only store lower triangle
  - Uses numbers > 1 for multiple edges
  - Or: a_{ij} stores edge weight
    - Use sentinel value like 0 or Inf for missing edges
  - Edge/vertex labels stored separately

Common graph operations

- **Graph(n)**
  - Initializes a graph with n vertices and 0 edges
- **AddEdge(u, v)**
  - Adds an edge from u to v
- **RemoveEdge(u, v)**
  - Removes the edge (u, v) from the graph
- **IsAdjacent(u, v)**
  - Returns whether (u, v) is an edge of the graph
- **GetNeighbors(v)**
  - Returns set of neighbors of v

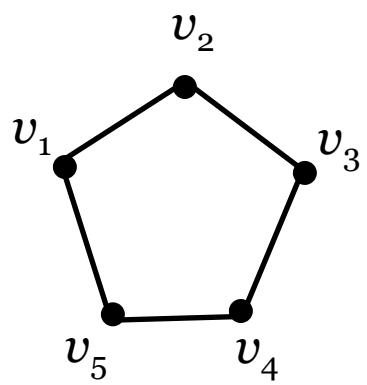
Adjacency matrix operations

- **Graph(n)**
  - Initialize n by n matrix
  - $\Theta(n^2)$ time
- **AddEdge(u, v)**
  - $a_{uv} = 1$
  - $\Theta(1)$ time
- **RemoveEdge(u, v)**
  - $a_{uv} = 0$
  - $\Theta(1)$ time
- **IsAdjacent(u, v)**
  - Return $a_{uv} = 1$
  - $\Theta(1)$ time
- **GetNeighbors(v)**
  - Scan row v of matrix and add u to list if $a_{vu} = 1$
  - $\Theta(n)$ time

Graph representations, cont'd

- **Adjacency list**
  - Stores list (or set) of neighbors for each vertex

v_1:	2	5
v_2:	1	3
v_3:	2	4
v_4:	3	5
v_5:	1	4



  - Neighbors are often sorted
    - Sometimes use hash table for large graphs
  - Can store edge weights or labels in `struct`/object
  - Can also store edges using a linked structure
    - Improves performance of adding/deleting vertices

Hash-based adjacency list

- **Main idea**
  - Store neighbors in hash table
- Graph: make empty hash tables
- Add neighbors: insert
- Remove: delete
- IsAdjacent: search
- GetNeighbors: iterate

Operation	Adjacency list
Graph(n)	
AddEdge(u, v)	
RemoveEdge(u, v)	
IsAdjacent(u, v)	
GetNeighbors(v)	

* Expected case
† Amortized

Hash-based adjacency list

- **Main idea**
  - Store neighbors in hash table
- Graph: make empty hash tables
- Add neighbors: insert
- Remove: delete
- IsAdjacent: search
- GetNeighbors: iterate

Operation	Adjacency list
Graph(n)	$\Theta(n)$
AddEdge(u, v)	$\Theta(1)$*†
RemoveEdge(u, v)	$\Theta(1)$*
IsAdjacent(u, v)	$\Theta(1)$*
GetNeighbors(v)	$\Theta(\deg(v))$

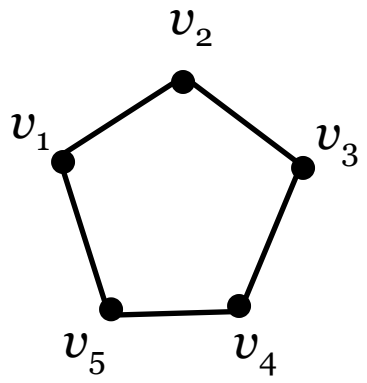
* Expected case
† Amortized

- Reduces expected complexity
- GetNeighbors still linear
- More complex
- Higher coefficients
- Poor worst-case performance

Graph representations, cont'd

- **Edge list**
  - List of all edges in graph
    - Usually sorted
    - Undirected: may or may not include "reverse" edges

1	2
1	5
2	3
3	4
4	5



  - Edge weights or labels appear after vertex IDs
  - Order and vertex labels represented separately

Graph representation analysis

Operation	Adjacency matrix	Adjacency list	Edge list
Graph(n)	$\Theta(n^2)$	$\Theta(n)$	$\Theta(1)$
AddEdge(u, v)	$\Theta(1)$	$\Theta(1)^*$	$\Theta(m)$
RemoveEdge(u, v)	$\Theta(1)$	$\Theta(1)^*$	$\Theta(m)$
IsAdjacent(u, v)	$\Theta(1)$	$\Theta(1)^*$	$\Theta(\lg(m))$
GetNeighbors(v)	$\Theta(n)$	$\Theta(\deg(v))$	$\Theta(m)$
Convert from:	**Adjacency matrix**	**Adjacency list**	**Edge list**
Adj. matrix	n/a	$\Theta(n^2)$	$\Theta(n^2)$
Adj. list	$\Theta(n^2)$	n/a	$\Theta(m)$
Edge list	$\Theta(n^2)$	$\Theta(m)$	n/a
Space:	$\Theta(n^2)$	$\Theta(m+n)$	$\Theta(m)$

- Matrix: good for small graphs or dense graphs
  - *Dense*: significant fraction of edges exist, $m = \Theta(n^2)$
- Adj. list: good for sparse graphs
  - Most "real world" graphs are sparse
- Edge list: commonly used to store on disk

Facebook friend graph statistics

- As of May 2011:
  - n: ~721 million
  - m: ~68.7 billion
  - Δ: 5000 (hard cap)
  - Avg degree: ~191
  - Median degree: 99
  - 99.91% are connected together
  - 99.6% are within 6 "hops" of one another
  - 92% are within 5 "hops"
- *Source*: "The Anatomy of the Facebook Social Graph," J. Ugander et al., arXiv.org, Nov 2011.

Very sparse!

Facebook graph representation

Average case timing (approx.):

Best choice!

Operation	Adjacency matrix	Adjacency list	Edge list
Graph(n)	~ 2 years	721 ms	1 ns
AddEdge(u, v)	1 ns	2 ns	68.7 s
RemoveEdge(u, v)	1 ns	2 ns	68.7 s
IsAdjacent(u, v)	1 ns	2 ns	36 ns
GetNeighbors(v)	721 ms	191 ns	68.7 s
Convert from:	**Adjacency matrix**	**Adjacency list**	**Edge list**
Adj. matrix	n/a	~ 2 years	~ 2 years
Adj. list	~ 2 years	n/a	68.7 s
Edge list	~ 2 years	68.7 s	n/a
Space:	~ 58 PB (million GB)	~517 GB	~512 GB

Graph algorithm analysis

- Analysis similar to general algorithm analysis
- Complexity given in terms of graph features
  - n: order (vertices)
  - m: size (edges)
  - $\deg(v)$: all neighbors of a given vertex
  - Δ: max degree
- Also depends on graph representation
- Sometimes useful to add up function calls separately
- **Example**

Dominance relationships

$$\deg(v) = O(\Delta),\ O(n),\ O(m)$$
$$\Delta = O(n),\ O(m)$$
$$m = O(n^2)$$

No direct relationship b/w n and m

```
Input: G = (V, E): a graph
Input: n: the number of vertices in G
Input: m: the number of edges in G
Output: the average degree of the vertices in G
Algorithm: AvgDegree
sum = 0
for v in V do
    for u in N(v) do
        sum = sum + 1
    end
end
return sum/n
```

Graph algorithm analysis

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- **Example**

Dominance relationships

$$\deg(v) = O(\Delta),\ O(n),\ O(m)$$
$$\Delta = O(n),\ O(m)$$
$$m = O(n^2)$$

No direct relationship b/w n and m

Cost to calculate $N(v)$:
$\Theta(\deg(v))$ for list
$\Theta(n)$ for matrix

Total:
$\sum_{v \in V} \Theta(\deg(v)) = \Theta(m)$ for list
$\Theta(n^2)$ for matrix

```
Input: G = (V, E): a graph
Input: n: the number of vertices in G
Input: m: the number of edges in G
Output: the average degree of the vertices in G
Algorithm: AvgDegree
sum = 0                          Θ(1)
for v in V do                    n iterations
    for u in N(v) do             Θ(deg(v)) iterations
        sum = sum + 1            Θ(1)
    end
end
return sum/n                     Θ(1)
```

Graph theory example

- What is the worst-case complexity for the following algorithm to compute the average degree in a graph?

Input: $G = (V, E)$: a graph
Input: n: the number of vertices in G
Input: m: the number of edges in G
Output: the average degree of the vertices in G
Algorithm: FastAverage
return $2m/n$

Graph theory example

- What is the worst-case complexity for the following algorithm to compute the average degree in a graph?

Input: $G = (V, E)$: a graph
Input: n: the number of vertices in G
Input: m: the number of edges in G
Output: the average degree of the vertices in G
Algorithm: FastAverage
return $2m/n$

- **Moral of the story:** knowledge is power

Graph algorithm exercise

```
Input: G = (V, E): graph with n vertices and m edges
Input: n, m: order and size of G
Output: number of triangles in G; that is, sets of 3
        vertices that are all adjacent
Algorithm: Triangles
tri = 0
for u = 1 to n do
    for v ∈ N(u) do
        for w = 1 to n do
            if w is adjacent to u and v then
                tri = tri + 1
            end
        end
    end
end
return tri/6
```

Operation	Adj. matrix	Opt. adj. list
IsAdjacent(u, v)	$\Theta(1)$	$\Theta(1)$*
GetNeighbors(v)	$\Theta(n)$	$\Theta(\deg(v))$*

1. What is the worst case time complexity for Triangles when using an adjacency matrix?
2. What is the expected case time complexity for Triangles when using an adjacency list with hash tables?
3. How does this algorithm compare with the naïve algorithm of checking all sets of 3 vertices individually?

Graph algorithm exercise

Input: $G = (V, E)$: graph with n vertices and m edges
Input: n, m: order and size of G
Output: number of *triangles* in G; that is, sets of 3 vertices that are all adjacent

Adj. matrix

Adj. list

$\Theta(1)$	1 **Algorithm:** Triangles
$\Theta(1)$	2 $tri = 0$
n iters	3 **for** $u = 1$ to n **do**
$\deg(u)$ iters, $\Theta(n)$	4 **for** $v \in N(u)$ **do**
n iters	5 **for** $w = 1$ to n **do**
$\Theta(1)$	6 **if** w is adjacent to u and v **then**
$\Theta(1)$	7 $tri = tri + 1$
	8 **end**
Lines 5–9: $\Theta(n)$	9 **end**
Lines 4–10: $\Theta(n \deg(u))$	10 **end**
Lines 3–11: $\sum_{u=1}^{n} \Theta(n \deg(u)) = \Theta(nm)$	11 **end**
$\Theta(1)$	12 **return** $tri/6$

$\Theta(\deg(u))$ for GetNeighbors

Everything else the same

Total time: $\Theta(nm)$

Graphs have $\binom{n}{3} = \Theta(n^3)$ sets of 3 vertices
This is strictly worse than $\Theta(nm)$

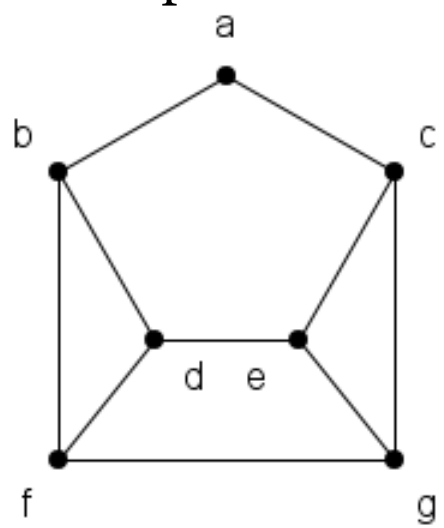
Total time: $\Theta(nm)$

Greedy algorithms review

- **Strategy:**
  - Break down problem into sequence of decisions
  - Make "best" choice for each decision
- **Example:** find largest subset with sum below a threshold
    - Choose min element
      - *Greedy idea:* min element gives us more "room" to find other elements
    - Repeat until next element would exceed threshold
- Proof of correctness is tricky
  - *Intuition:* greedy is only correct when greedy choice always "as least as good" as any alternative
  - Many greedy algorithms are incorrect
- Natural choice for optimization problems
- Very efficient (heaps!)
- Can be used as a heuristic
- Often not correct

Graph traversal terminology

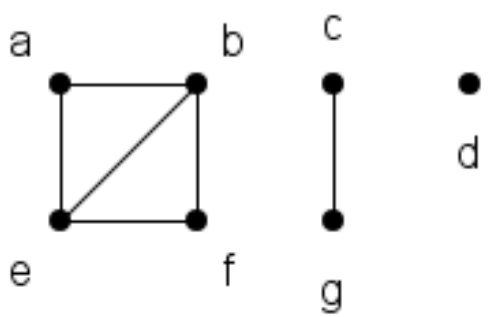
- **Walk:** sequence of vertices joined by edges
  - If directed, must obey directionality of edges
  - May include the same vertex or edge multiple times
  - **Length of a walk:** number of edges (vertices – 1)
    - If weighted, sum of edge weights
- **Path:** walk with no repeated vertices or edges
- **Circuit:** walk that begins and ends at the same vertex
  - A.k.a., "closed walk"
- **Cycle:** nontrivial path that begins and ends at the same vertex
- **Example**



Walk: b, d, e, g, f, g, c
Path: b, a, c, e, d, f, g
Circuit: f, d, e, c, a, b, d, e, g, f
Cycle: b, a, c, g, f, b

Graph connectivity terminology

- **Connected:** two vertices that have a path between them
  - Also, a graph in which all vertices are connected
  - **Theorem:** if there exists a (u, v)-walk, there exists a (u, v)-path
- **Component:** maximal set of connected vertices in a graph
  - Maximal = cannot be enlarged by adding more vertices
- **Distance:** length of the shortest path between two vertices
  - Only defined for connected vertices
  - Denoted d(u, v) or $d_G(u, v)$
  - Positive definite, symmetric (if undirected), triangle inequality
- **Example**



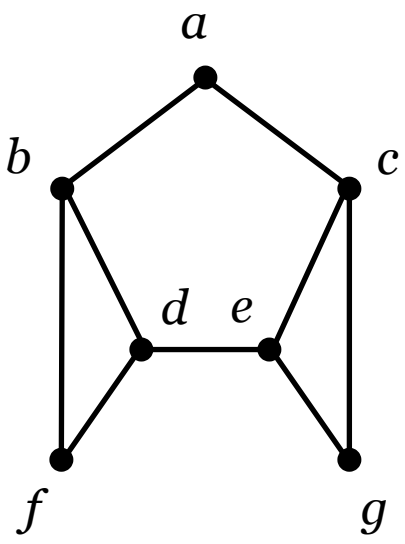
Connected: a and f (not adjacent!)
Components: {a, b, e, f}, {c, g}, {d}
Distance: d(a, f) = 2

Graph traversals

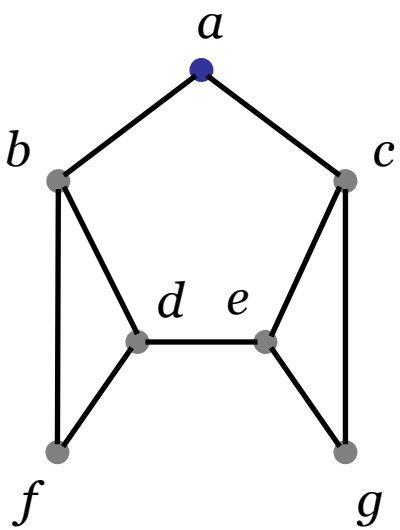
- **Traversal:** iterating through graph in connected order
  - Basis for many graph algorithms
  - Start at a specific vertex, iterate through all connected vertices
- Two main strategies: breadth-first and depth-first traversal
  - Both mark vertices in order as they go
- **Breadth-first search (BFS)**
  - Mark all vertices as undiscovered
  - Add v_o to queue
  - Mark v_o as discovered
  - While queue is not empty:
    - Dequeue vertex v
    - Enqueue children of v that are undiscovered
    - Mark v as explored
    - Mark children as discovered
  - If any vertex still undiscovered, choose one and restart

BFS example

- In what order would BFS process the vertices of the graph below when starting at vertex a?



BFS example

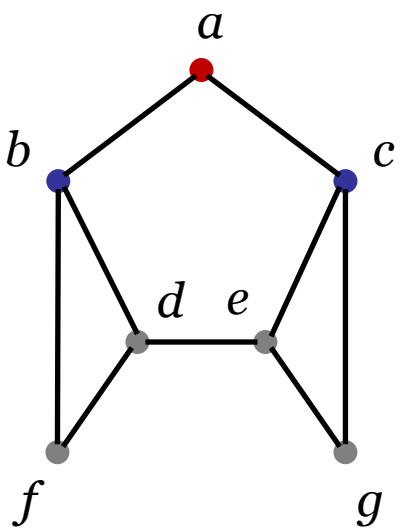


Undiscovered
Discovered
Explored

Queue: a

Traversal order:

BFS example

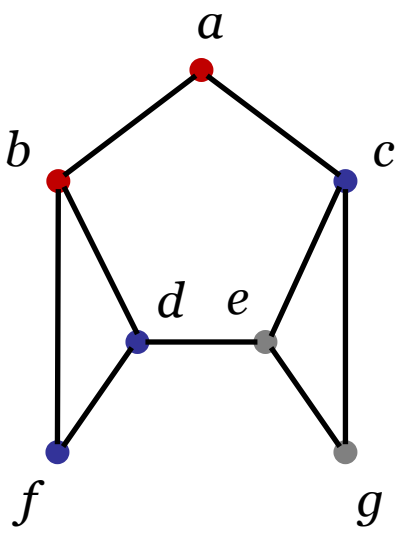


Undiscovered
Discovered
Explored

Queue: *b, c*

Traversal order: *a*

BFS example



Undiscovered
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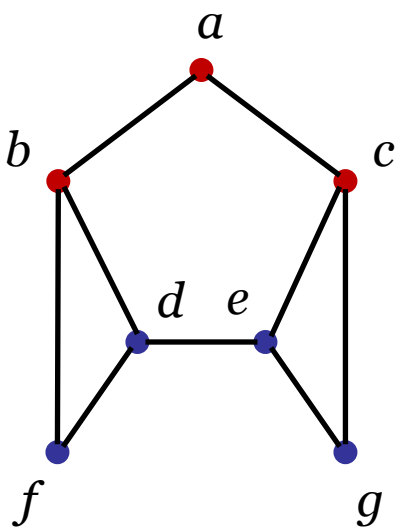
Queue: c, d, f

Traversal order: a, b

BFS example



Undiscovered
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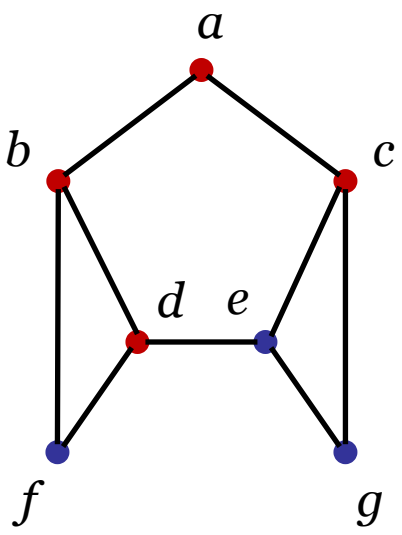
Explored

Queue: *d, f, e, g*

Traversal order: *a, b, c*

BFS example

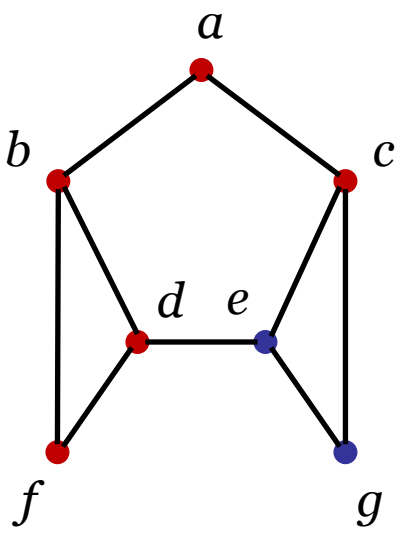


Undiscovered
Discovered
Explored

Queue: *f, e, g*

Traversal order: *a, b, c, d*

BFS example

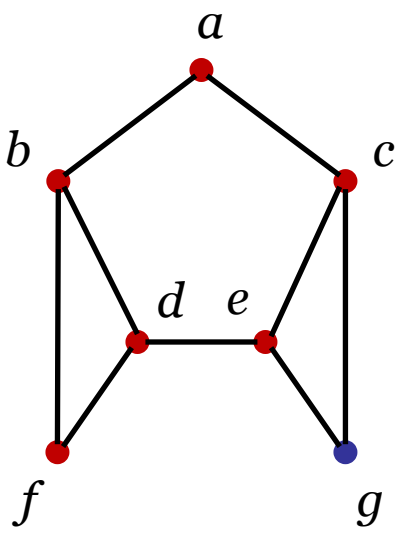


Undiscovered
Discovered
Explored

Queue: *e, g*

Traversal order: *a, b, c, d, f*

BFS example

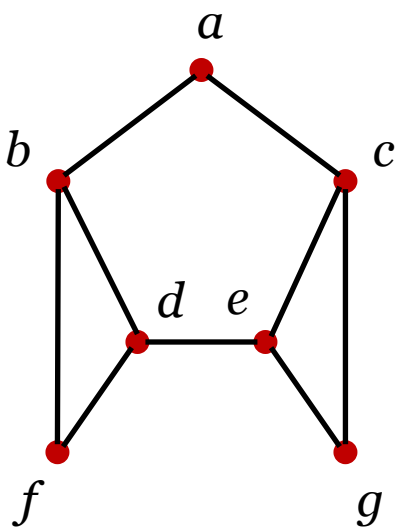


Undiscovered
Discovered
Explored

Queue: *g*

Traversal order: *a, b, c, d, f, e*

BFS example



Undiscovered
Discovered
Explored

Queue:

Traversal order: *a, b, c, d, f, e, g*

Depth-first search

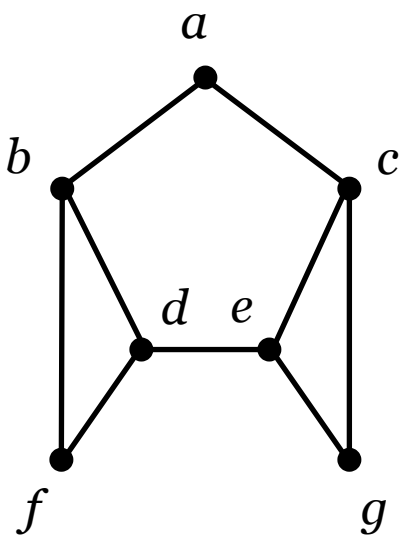
- Recursive algorithm
  - Simpler code than BFS
  - Implicitly stack-based

- DFS:
  - Call DFS-Recursive(v_0)
  - If any vertex still undiscovered, choose one and restart
- DFS-Recursive(v):
  - Mark v as discovered
  - For c in N(v):
    - If c is unexplored
      - Call DFS-Recursive(c)
  - Mark v as explored

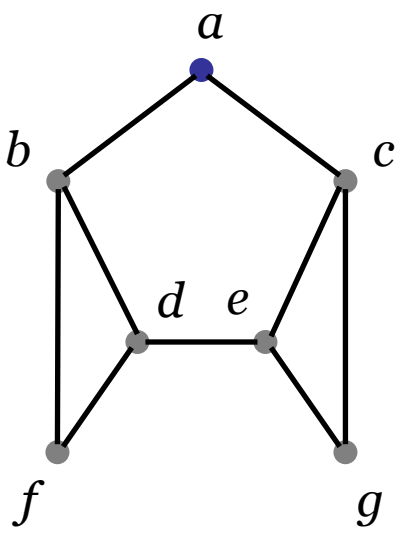
- **Observation:** vertices are discovered iff they are in stack/queue

DFS example

- In what order would DFS process the vertices of the graph below when starting at vertex a?



DFS example

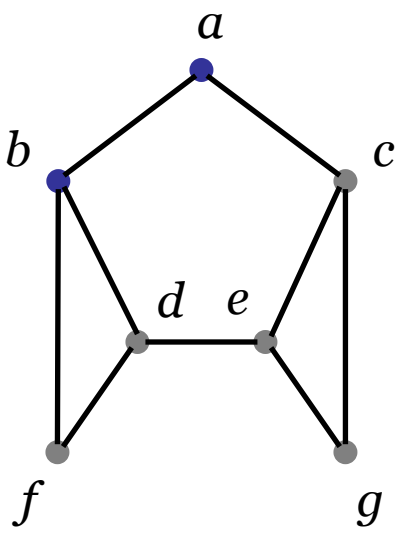


Undiscovered
Discovered
Explored

Call stack: a

Traversal order:

DFS example

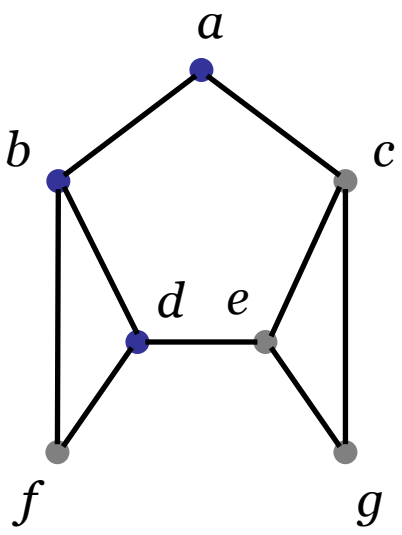


Undiscovered
Discovered
Explored

Call stack: *a*, *b*

Traversal order: *a*

DFS example

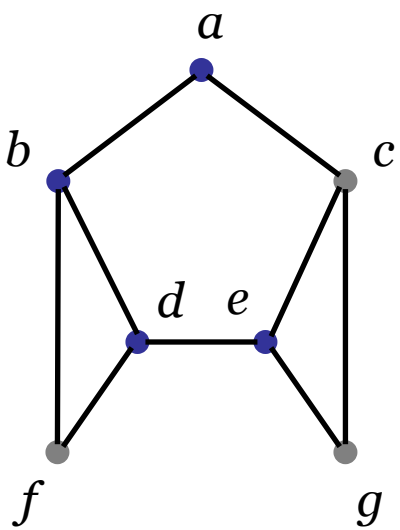


Undiscovered
Discovered
Explored

Call stack: *a*, *b*, *d*

Traversal order: *a*, *b*

DFS example

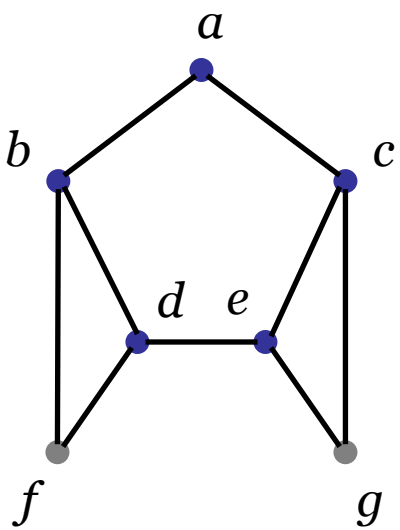


Undiscovered
Discovered
Explored

Call stack: *a, b, d, e*

Traversal order: *a, b, d*

DFS example

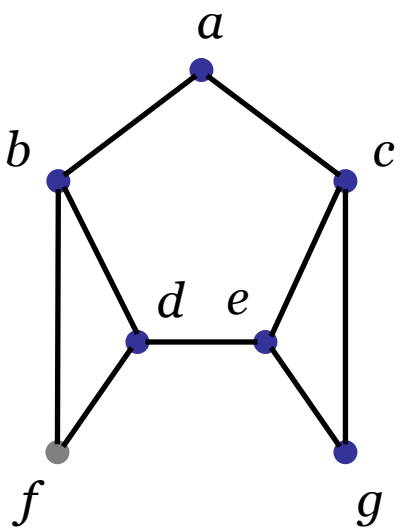


Undiscovered
Discovered
Explored

Call stack: *a, b, d, e, c*

Traversal order: *a, b, d, e*

DFS example

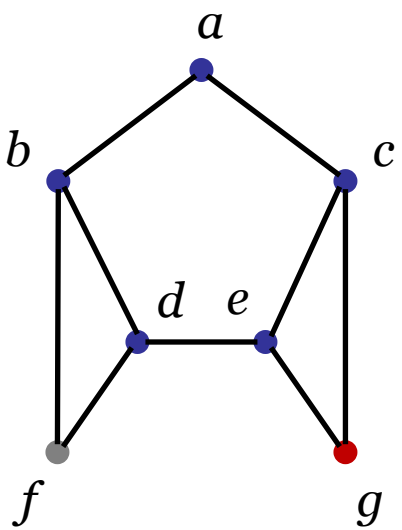


Undiscovered
Discovered
Explored

Call stack: *a, b, d, e, c, g*

Traversal order: *a, b, d, e, c*

DFS example

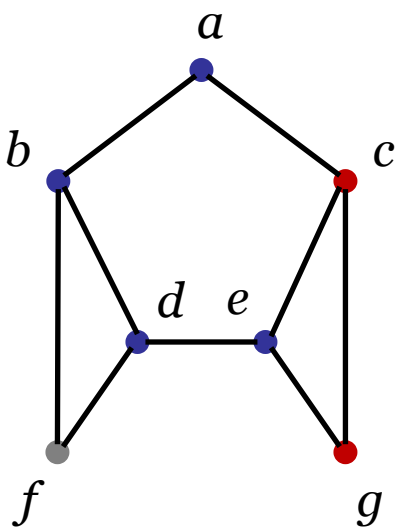


Undiscovered
Discovered
Explored

Call stack: *a, b, d, e, c*

Traversal order: *a, b, d, e, c, g*

DFS example

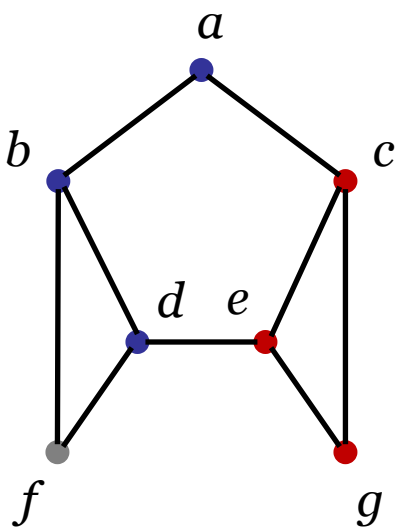


Undiscovered
Discovered
Explored

Call stack: *a, b, d, e*

Traversal order: *a, b, d, e, c, g*

DFS example

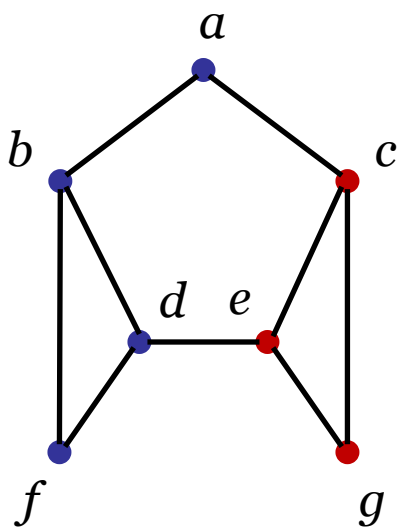


Undiscovered
Discovered
Explored

Call stack: *a, b, d*

Traversal order: *a, b, d, e, c, g*

DFS example

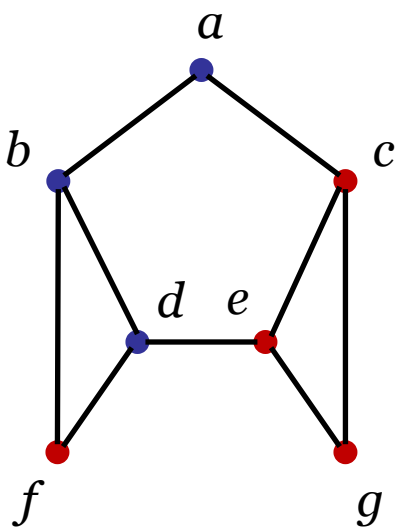


Undiscovered
Discovered
Explored

Call stack: *a, b, d, f*

Traversal order: *a, b, d, e, c, g, f*

DFS example

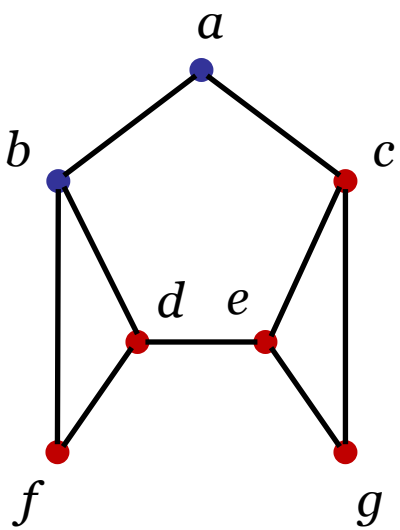


Undiscovered
Discovered
Explored

Call stack: *a, b, d*

Traversal order: *a, b, d, e, c, g, f*

DFS example



Undiscovered
Discovered
Explored

Call stack: *a, b*

Traversal order: *a, b, d, e, c, g, f*

Traversal complexity example

- What is the complexity of BFS with an adjacency list? Adjacency matrix?
  - How many times do we find the neighbors of a vertex?
  - How many total times do we increment count?

```
Input: G = (V, E): input graph with n vertices
       and m edges
Input: m, n: size and order of G
Mark all vertices of G as undiscovered
Let q be a new queue
count = 1
repeat
    Add V[count] to q
    while q ≠ ∅ do
        Get next vertex from q
        /* Process this vertex                    */
        Mark this vertex as explored
        Add all undiscovered neighbors of this
         vertex to q and mark as discovered
    end
    Increase count until V[count] is undiscovered
until count > n
return
```

Traversal complexity example

- What is the complexity of BFS with an adjacency list? Adjacency matrix?
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```
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until count > n
return
```

Adjacency list:

- Mark all vertices: $\Theta(n)$
- Loop iterates once per vertex: $\Theta(n)$
- Call getNeighbors on all vertices once: $\Theta(m)$
- *count* incremented n times: $\Theta(n)$
- ***Total:*** $\Theta(n+m)$

Adjacency matrix:

- Mark all vertices: $\Theta(n)$
- Loop iterates once per vertex: $\Theta(n)$
- Call getNeighbors on all vertices once: $\Theta(n^2)$
- *count* incremented n times: $\Theta(n)$
- ***Total:*** $\Theta(n^2)$

Coming up

- Traversal-based algorithms
- Weighted graph algorithms

- **Recommended readings:** Sections 13.1-13.4
- *Practice problems:* R-13.1, R-13.2, R-13.5, R-13.7, C-13.4